

Testing Mechanisms Across the Software Life Cycle

Functional Testing • System
Integration • User Testing • Platform
Testing • Testing Tools

Introduction

- Testing mechanisms are essential across all stages of the software development life cycle (SDLC). They ensure the software meets functional requirements, performs reliably, and delivers a good user experience.
- Key testing stages include:
 - Functional Testing
 - System Integration Testing
 - User Testing
 - Platform/Compatibility Testing

Functional Testing

- Objective: Verify that the software performs each function as specified.
- Conducted using requirements and design documents.
- Involves testing features, APIs, and user interactions.
- Types: Unit testing, smoke testing, sanity testing.
- Tools: Selenium, JUnit, TestNG, Postman.

System Integration Testing

- Objective: Ensure that integrated modules or subsystems work together correctly.
- Validates data flow between modules.
- Detects interface and communication errors.
- Often performed after unit testing.
- Tools: Jenkins, Apache JMeter, SoapUI, Docker.

User Testing (UAT – User Acceptance Testing)

- Objective: Confirm that the software meets end-user needs and business requirements.
- Conducted in a real-world or simulated environment.
- Involves actual users performing defined tasks.
- Ensures usability, satisfaction, and readiness for deployment.
- Tools: TestRail, Zephyr, Bugzilla.

Testing on Different Platforms

- Objective: Verify software performance across various operating systems, devices, and browsers.
- Ensures consistent behavior across environments.
- Important for mobile and web applications.
- Tests include OS compatibility, browser compatibility, and device testing.
- Tools: BrowserStack, Sauce Labs, Appium.

Testing Tools Overview

- Common Testing Tools:
- Selenium – Web automation testing
- JUnit/TestNG – Unit testing for Java
- Postman – API testing
- JMeter – Performance and load testing
- Jenkins – Continuous integration testing
- Appium – Mobile application testing
- BrowserStack – Cross-platform compatibility testing